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Free Choice Readings of Disjunctive Sentences and Implicatures of Existence*

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Jae-II Yeom (2013), Free Choice Readings of Disjunctive Sentences and Implicatures of Existence. *Journal of Language Sciences* 20(1), 243–264. There are cases where a disjunction sentence reads like the corresponding conjunction sentence. It is called a Free Choice Effect(= FCE). Simons(2005), Klindinst(2006) and Fox(2006) attempted to account for the effect, but they are not fully successful. In this paper, I employ the positive features of their analyses and avoid the problems with them. Simons's analysis implies that each disjunct makes a separate meaning contribution. No syntactic analyses like Klindinst's or Fox's, which take FCEs as part of scalar implicatures, can account for FCEs. I claim that in a disjunction structure, each disjunct makes a separate meaning contribution to a quantifier which has wide scope over the disjunction structure. I call them implicatures of existence. The implicatures should in turn contribute to the meaning of the whole sentence. (Hongik University)

Key Words: disjunction, Free Choice Effect, implicature of existence, scalar implicature, meaning contribution, conjunction

1. Introduction

Kamp(1973, 1979) observed that some disjunction sentences are interpreted like a conjunction sentence. The first case that caught his attention was a permission statement with a disjunction structure:

(1) You may eat an apple or a pear.

+> You may eat an apple & you may eat a pear. ('+>': implicates)

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In general, a statement with or makes a weaker statement than an alternative statement with only one disjunct, but in (1) the opposite seems to hold. This meaning relationship is not limited to permission statements:

- (2) Some students were sleeping or watching TV.

+> Some students were sleeping.

+> Some students were watching TV.

- (3) John might be in the US or in Germany.

+> John might be in the US.

+> John might be in Germany.

One common factor is that an existential quantifier has wide scope over a disjunction structure. In (1), (2) and (3), may, some and might are existential quantifiers over deontic alternatives of the actual world, individuals in the actual world, and epistemic alternatives of the actual world, respectively. Such an inference pattern is called a Free Choice Effect(FCE), which can be described as follows:

- (4) General Pattern of a FC interpretation:

$\phi(A \text{ OR } B) \rightarrow \phi(A) \text{ AND } \phi(B)$, where

ϕ involves an existential operator which has wide scope over or.

A FCE has not been explained as a standard scalar implicature(= SI).

The basic principle underlying SIs is Grice's maxim of quantity, but Grice(1975, 1989) gave no precise procedure of deriving a SI. Following his basic idea, Sauerland(2004) proposed a more precise procedure.

- (5) I $\psi \in \text{ScalAlt}(\phi)$ and $\psi \Rightarrow \phi$ and not $\phi \Rightarrow \psi$, then $\neg K\psi$ is a primary implicature (= PrI) of ϕ . (Ignorance Inferences)

If $\neg K\psi$ is a PrI of ϕ and $K\neg\psi$ is consistent with the conjunction of ϕ and all PrIs of ϕ , then $K\neg\psi$ becomes a secondary implicature (= SeI) of ϕ .

A stronger scalar alternative ψ of an asserted sentence ϕ leads to an implicature with the negation operator in it stepwise. Grice's maxim of quantity only predicts the "not-know" (= " $\neg K$ ") implicatures $\neg K\psi$, which Sauerland calls a primary implicature (= PrI). The PrI is rather trivial and the hearer tentatively strengthens the PrI into the "know-not" (= " $K\neg$ ") implicature $K\neg\psi$ and sees if it is compatible with the rest of the PrIs plus the assertion. If it is, then the hearer concludes that the stronger implicature is what the speaker intends. Sauerland calls the " $K\neg$ " implicature a secondary implicature (= SeI).

This procedure does not explain the FC interpretations observed in (1), (2) and (3). Sauerland assumes the alternative set of 'A or B' is {A and B, A, B, A or B}.

(6) PrIs:

- i. $\neg K$ (Some students were sleeping and watching TV)
- ii. $\neg K$ (Some students were sleeping)
- iii. $\neg K$ (Some students were watching TV)

SeI:

$K\neg$ (Some students were sleeping and watching TV)

FCEs:

- i. $K\neg$ (Some students were sleeping)
- ii. $K\neg$ (Some students were watching TV)

Among the PrIs, PrI (i) becomes a SeI, but PrIs (ii) and (iii) do not. If PrI (ii) did, it would be incompatible with PrI (iii), and vice versa. Moreover, PrI (ii) and PrI (iii) are in conflict with the FCEs we want. Sauerland's approach simply predicts that FCEs cannot be derived as SeIs.

This indicates that FCEs are not derived on the basis of Grice's (or Sauerland's) maxim of quantity. We need a new approach to account for them. In this paper I claim that FCEs are conditions for each disjunct to be used non-trivially. This is based on the general principle that an expression in a sentence must contribute to the meaning of the sentence. The paper is organized as follows: in Section 2, I will review some

previous analyses of FCEs critically. In Section 3, I claim that FCEs are implicatures of existence, on the basis of more extensive observations. Each disjunct should make an exclusive meaning contribution to the interpretation of a quantifier which has wide scope over the disjunction structure. In Section 4, I show that FCEs cannot be dealt with syntactically. In Section 5, I will explain how FCEs are dealt with in my analysis. FCEs are basically calculated locally, but the effects must contribute to the meaning of a sentence as a whole. In Section 6, I will discuss how locally derived FCEs contribute to the meaning of a whole sentence. In Section 7, I summarize the discussions in the paper.

2. Properties of FECs in Previous Analyses

Recently there have been some attempts to explain how FCEs arise, including Simons(2005), Fox(2006), Klinedinst(2006), Eckardt(2007), etc. In this section I will go over the properties of FECs by reviewing the first three.

2.1. Simons(1998, 2005)

In Simons's(2005) analysis, each disjunct is interpreted with the rest of the sentence, independently of the other disjunct(s).¹⁾ It gives rise to a set of interpretations and an operator exploits the set to yield the truth-conditions of the sentence.²⁾ Much of the paper is devoted to the interpretation rules for independent compositional interpretation. First, a disjunction structure is interpreted as a set of meanings of disjuncts:

(7) Interpretation rule for or coordinations

$$\| [a_1 \text{ or } \dots \text{ or } a_n] \| = \{ \| a_1 \|, \dots, \| a_n \| \}$$

1) Aloni(2007) also takes a similar strategy of independent interpretation of each disjunct.

2) Larson(1985) and Schwarz(1999) attempt to capture the scope of the disjunction operator, but Simons(2005) points out some problem with such syntactic approaches.

And the meaning of each disjunct is compositionally interpreted with the rest of the sentence, independently of the other(s), still giving rise to a set of interpretations:

(8) Rule of Independent Composition(optional)

- a. Let α be a branching node with daughters β and γ , where

$$\|\beta\| \in D_{\langle b,a \rangle} \text{ and } \|\gamma\| \subseteq D_b.$$

$$\text{Then } \|\alpha\| = \{a: \exists g \in \|\gamma\|. \|\beta\|(g) = a\}.$$

- b. Let α be a branching node with daughters β and γ , where

$$\|\beta\| \subseteq D_{\langle b,a \rangle} \text{ and } \|\gamma\| \in D_b.$$

$$\text{Then } \|\alpha\| = \{a: \exists b \in \|\beta\|. \|\gamma\|(b) = a\}.$$

- c. Let α be a branching node with daughters β and γ , where

$$\|\beta\| \subseteq D_b \text{ and } \|\gamma\| \subseteq D_{\langle b,a \rangle}.$$

$$\text{Then } \|\alpha\| = \{a: \exists g \in \|\gamma\|, b \in \|\beta\|. g(b)=a\}.$$

[Notation: for any function f and set X , let $f[X] = \{f(x): x \in X\}$.]

- d. Let α be a branching node with daughters β and γ , where

$$\|\beta\| \subseteq D_b \text{ and } \|\gamma\| \subseteq D_{\langle b,a \rangle}.$$

$$\text{Then } \|\alpha\| = \{S: \exists g \in \|\gamma\|. S = g[\|\beta\|]\}.$$

In (8a), if a constituent α has two daughters β and γ and the meaning of β is a set of interpretations, then each member of the set is combined with the meaning of γ . The meaning of γ could be a set of interpretations, and in such a case rule (8b) applies. If the meanings of both β and γ are sets of interpretations, rule (8c) applies. If the subject is a disjunction structure and it has wide scope over the predicate, then rule (8d) applies. One simple example is given here:

$$(9) \|\text{sing or dance}\| = \{\|\text{sing}\|, \|\text{dance}\|\}$$

$$\|\text{John sings or dances}\| = \{\|\text{sing}(j)\|, \|\text{dance}(j)\|\}$$

Independent composition stops at some point of interpretation, when there is a quantifier, and each alternative is incorporated in the truth conditions by being supercovered:

- (10) A non-empty set SC is a supercover of S iff
 every member of SC contains some member of S;
 every member of S belongs to some member of SC.
 If SC is a supercover of S, then $\cup SC \supseteq S$.

Using the notion of supercover, the truth conditions of John sings or dances is the following:

- (11) $\exists S . w^* \in S \ \& \ \{ \parallel \text{sing}(j) \parallel , \parallel \text{dance}(j) \parallel \}$ is a supercover of S.
 (w*: actual world)

As Simons mentioned, Winter(1995) also proposes to interpret a conjunction structure with each conjunct interpreted, independently of the other conjunct(s). Simons claims that this supports his analysis. However, if a conjunction structure and a disjunction structure are mixed, we do not know which alternative comes from which coordination operator.

- (12) It might be the case that John sang or played a musical instrument and that Mary danced.
 a. $\{ \parallel \text{John sang} \parallel , \parallel \text{John played a mus. inst.} \parallel , \parallel \text{Mary danced} \parallel \}$
 b. $\{ \{ \parallel \text{John sang} \parallel , \parallel \text{John played a mus. inst.} \parallel \} , \parallel \text{Mary danced} \parallel \}$

Until might is interpreted, each coordinate is interpreted independently of the others. I suppose that there are two ways of dealing with the two coordinates. One is to interpret all coordinates pointwise, as in (12a), and the other to interpret a disjunction structure and a conjunct structure in a way that one structure is embedded in the other, as in (12b). Which way we may take, we cannot tell which set comes from a disjunction structure or a conjunction structure when might is interpreted. Simons also mentions the alternative semantics of focus phenomena as another motivation for his analysis. This will make the picture more complex.

Despite this problem, Simons's and Winter's analyses imply that each conjunct or disjunct can and should contribute to the meaning of the

whole sentence, independently of the other conjunct(s) or disjunct(s).

2.2. Fox(2006)

Fox(2006) accepts Sauerland's(2004) idea of SeIs as strengthened PrIs. He assumes, following Sauerland(2004), that the alternative set of or is {and, R, L, or}. "A R B" and "A L B" are equivalent to B and A, respectively. He assumes a covert syntactic operator Exh(austive operator) which negates all non-weaker alternatives which are innocently excludable, given an alternative set.³⁾ This is similar to Sauerland's idea of PrIs being strengthened into SeIs. An alternative q is innocently excludable given A unless the negation of q requires a stronger alternative to be true.

- (13) $\parallel \text{Exh} \parallel (A_{\langle \text{st}, \text{t} \rangle})(p_{\text{st}})(w) \Leftrightarrow p(w) \ \& \ \forall q \in \text{NW}(p, A) [q \text{ is innoc(ently) exclud(able) given } A \rightarrow \neg q(w)]$
 q is innoc. exclud. given A if $\neg \exists q' \in \text{NW}(p, A) [[p \wedge \neg q] \Rightarrow q']$
 p: a sentence which follows the Exh operator
 A: a set of alternatives
 NW(p,A): a set of non-weaker alternatives than p in A

The Exh operator takes a set of alternatives A and an utterance p, and attempts to exclude as many propositions from A in a way that would be consistent with the requirement that the utterance be true.

Fox also accepts Kratzer and Shimoyama's(2002) anti-exhaustivity principle. In calculating SIs, he compares the meanings of alternatives strengthened by their implicatures, not the literal meanings. To get this effect, he applies the Exh recursively.

- (14) Recursive Parsing-Strategy: If a sentence S has an undesirable Ignorance Inference, parse it as $\text{Exh}(\text{Alt}(S))(S)$.

Consider a concrete example of p or q. By Grice's maxim of quantity, this

3) Fox follows Chierchia(2004) in taking a syntactic approach.

yields a variety of PrIs, which might be implausible in a particular context, and if so, would motivate the introduction of an exhaustive operator, $\text{Exh}(\text{Alt}(p \text{ or } q))(p \text{ or } q)$.

(15) You may eat the cake or the ice-cream.

p : you eat the cake

q : you eat the ice-cream

$\Diamond$: may

$C = \{\Diamond(p \vee q), \Diamond p, \Diamond q, \Diamond(p \wedge q)\}$

$\text{Exh}(C)(\Diamond(p \vee q)) = \Diamond(p \vee q) \wedge \neg \Diamond(p \wedge q)$.

$\Diamond(p \wedge q)$ is the only proposition in $\text{Alt}(\Diamond(p \vee q))$ that can be innocently excluded, given the set of alternatives C . If we excluding $\Diamond p$, we will necessarily include $\Diamond q$, and vice versa. Hence, the meaning of (15) in our modal logic formalization is $\Diamond(p \vee q) \wedge \neg \Diamond(p \wedge q)$.

This yields the PrI that the speaker does not know what the hearer may eat. Thus the hearer applies the parsing strategy again. C' is a set of the strengthened meanings of the alternatives of the original utterance.

(16) $\text{Exh}(C')[\text{Exh}(C)(\text{You may eat the cake or the ice-cream})]$ where

$C' = \{\text{Exh}(C)(p) : p \in C\} = \{1. \text{Exh}(C)(\Diamond(p \vee q)), 2. \text{Exh}(C)(\Diamond p),$
 3. $\text{Exh}(C)(\Diamond q), 4. \text{Exh}(C)(\Diamond(p \wedge q))\}$

a. $\text{Exh}(C)(\Diamond(p \vee q)) = \Diamond(p \vee q) \wedge \neg \Diamond(p \wedge q)$

(crucially $= \neg(\Diamond p \wedge \neg \Diamond q) \vee (\Diamond q \wedge \neg \Diamond p)$)

b. $\text{Exh}(C)(\Diamond p) = \Diamond p \wedge \neg \Diamond q$

c. $\text{Exh}(C)(\Diamond q) = \Diamond q \wedge \neg \Diamond p$

d. $\text{Exh}(C)(\Diamond(p \wedge q)) = \Diamond(p \wedge q)$ (already excluded in $\text{Exh}(C)$)

$C' = \{\Diamond(p \vee q) \wedge \neg \Diamond(p \wedge q), \Diamond p \wedge \neg \Diamond q, \Diamond q \wedge \neg \Diamond p\}$

Here $\Diamond p \wedge \neg \Diamond q$ and $\Diamond q \wedge \neg \Diamond p$ can be innocently excluded, in addition to $\neg \Diamond(p \wedge q)$, from (16). Hence we get the following:

$$\begin{aligned}
 (17) \text{Exc}(C')(\text{Exh}(C)(\Diamond(p \vee q))) = \\
 \text{i. } \Diamond(p \vee q) \wedge \neg \Diamond(p \wedge q) \text{ and} \\
 \text{ii. } \neg(\Diamond p \wedge \neg \Diamond q) \text{ and} \\
 \text{iii. } \neg(\Diamond q \wedge \neg \Diamond p) \\
 = \text{(a) } \Diamond(p) \wedge \Diamond(q) \text{ and (b) } \neg \Diamond(p \wedge q)
 \end{aligned}$$

Here (17iia) is the FCE from the use of the disjunction structure, and (17iiib) is the standard scalar implicature. That is, the utterance You may eat the cake or the ice-cream implicates that you may eat the cake and you may eat the ice-cream, and that you may not eat both.

One problem with Fox's(2006) analysis is with the second prediction that $\neg \Diamond(p \wedge q)$, which means the hearer may not eat both. This seems too strong. As Simons(2005) claims, Jane may eat the cake or the ice-cream does not implicate that Jane may not eat both. To remedy this problem, he additionally applies Exh to each of the two disjuncts assuming that they are focused and have the same set of alternatives.

$$(18) \text{Exh}(C'')(\text{Exh}(C')(\Diamond(\text{Exh}(C1)(p) \text{ or } \text{Exh}(C2)(q)))).$$

Here both disjuncts p and q are focused so that $C1$ (the alternative set of p) = $C2$ (the alternative set of q). From this, he gets the right result:

$$\begin{aligned}
 (19) \text{Exh}(C'')(\text{Exh}(C')(\Diamond(p! \vee q!))) \\
 = \Diamond(p! \vee q!) \wedge \neg \Diamond(p! \wedge q!) \wedge \Diamond(p!) \wedge \Diamond(q!) \\
 = \Diamond(p! \vee q!) \wedge \Diamond(p!) \wedge \Diamond(q!) \\
 (p! := \text{Exh}(C1)(p) = p \wedge \neg q, q! := \text{Exh}(C2)(q) = q \wedge \neg p)
 \end{aligned}$$

However, no focus is involved in such sentences and he does not explain why Exh applies to each disjunct.

Second, Fox(2006) starts from the Gricean maxim of quantity, which applies to whole utterances. Fox applies the Exh to a set of alternatives of a whole sentence. But when he runs into a problem of generating the implicature that $\neg \Diamond(p \wedge q)$, he also applies the Exh sentence-internally.

This is problematic because the Gricean maxim of quantity is basically supposed to apply to whole statements. He is getting away from the basic assumption he started with.

Third, Fox assumes that FCEs are SIs but there is some evidence that it is not the case.⁴⁾ A direct motivation for claiming that FCEs are SIs is Alonso-Ovalle's(2005) observation that FCEs do not arise in downward entailing(= DE) contexts, just like any other SI(Kratzer and Shimoyama, 2002; Alonso-Ovalle, 2005; Fox, 2006; Klindinst, 2006, etc.):

- (20) a. His mother doesn't think that John may eat an apple or a pear.
 b. His mother thinks that NOT [John may eat an apple or a pear]
 (Standard meaning)
 = His mother thinks that John is not allowed to eat an apple and that he is not allowed to eat a pear(either).
 c. His mother thinks that NOT [John may eat an apple and John may eat a pear] (Negation of the FC reading)
 = His mother thinks that(either) John is not allowed to eat an apple or John is not allowed to eat a pear.

The meaning of sentence (20a) is (20b), in which no FCE arises. If a FCE arose, it would be interpreted as in (20c). Alonso-Ovalle attributes this to the fact that the disjunction structure lies in a DE context.

A SI arises only if a scalar term in a sentence is replaced with a stronger alternative and the whole sentence gets stronger, that is, only if it is in an upward entailing(= UE) context. Thus we can predict that if a scalar term is used in a non-monotonic or DE context, it does not yield a SI. One such context is the antecedent clause of a conditional:

- (21) a. If some guests found the taste of the stew strange or got sick,
 the stew must have been spoiled. (Eckardt, 2007)
 b. ??If some guests found the taste of the stew strange or got
 sick but didn't do both, the stew must have been spoiled. (SI)

4) Sauerland(2009) also claims that FCEs belong to a different kind of implicatures, mentioning Chelma(2009).

- c. If some guests found the taste of the stew strange and some guests got sick, then the stew must have been spoiled.

In the antecedent clause, the use of *or* does not yield a SI. Thus (21a) is not understood as meaning (21b). On the other hand, we still get FCEs in the same context. So we can get the meaning of (21c). This shows that FCEs are not SIs. This casts doubt on Fox's analysis.

2.3. Klinedinst(2006)

Klinedinst(2006) proposes a syntactic analysis of FCEs on the basis of the neo-Gricean SI, assuming the null operators *R* and *L*. He claims that SIs arise only when the relevant quantifier is in plural form. A plural form introduces a variable for plural entities, and a plural entity introduces a universal quantifier over individuals that constitute the plural entity. Klinedinst introduces an operator "STRONG" for this universal quantifier. STRONG(*X*) is the neo-Gricean strengthening of *X*, by which *X* is conjoined with the negation of its stronger neo-Gricean scalemates. This leads to FECs, as given in (22ii) and (22iii):

- (22) STRONG($\forall x: Ax \vee Bx$) is thus equivalent to
 $(\forall x: Ax \vee Bx) \wedge \neg(\forall x: Ax) \wedge \neg(\forall x: Bx) \wedge \neg(\forall x: Ax \wedge Bx)$
 $= (i)(\forall x: Ax \vee Bx) \wedge (ii)(\exists x: Ax \wedge \neg Bx) \wedge (iii)(\exists x: Bx \wedge \neg Ax)$
- (23) Some of the students each bought a car or a motorcycle.
- $[\exists X: \text{Students}(X)] \text{STRONG}([\forall x: x \text{ is-part-of } X] Cx \vee Mx)$
 - $= [\exists X: \text{Students}(X)] ([\forall x: x \text{ is-part-of } X] Cx \vee Mx) \wedge \neg([\forall x: x \text{ is-part-of } X] Cx) \wedge \neg([\forall x: x \text{ is-part-of } X] Mx) \wedge \neg([\forall x: x \text{ is-part-of } X] Cx \wedge Mx)$
 - $= [\exists X: \text{Students}(X)] ([\forall x: x \text{ is-part-of } X] Cx \vee Mx) \wedge ([\exists x: x \text{ is-part-of } X] Cx \wedge \neg Mx) \wedge ([\exists x: x \text{ is-part-of } X] Mx \wedge \neg Cx)$
 - 'At least one of the students bought a car (but not a motorcycle), and at least one of the students bought a motorcycle (but not a car).'

In typical cases, Klinedinst makes correct predictions.

However, this analysis shares some problems with other analyses. First, he attempts to explain FCEs as SIs, but I have also shown that FCEs do arise even in non-monotonic contexts, as shown in (21). This is related to a second problem. The operator STRONG is introduced in a local context, regardless of the monotonicity of the context in which a disjunction structure occurs. Whether this may make correct predictions, it means that his analysis is not correct. If it does, it means that FCEs are different from SIs. If it does not, it means his analysis does not work.

Third, Klinedinst introduced the operator STRONG to account for FCEs, but it does not have to be introduced only to interpret a plural NP with existential force. And plurality might be an epi-phenomenon.

- (24) a. Every student sang or danced.
b. Most students sang or danced.

In (24a) we have a singular NP, but in (24b) we have a plural NP. Whether the subject is plural or not, we can infer that some students sang and did not dance and some danced but did not sing. It is exactly what we infer as a FCE when the NP is an existential quantifier. But the two sentences are interpreted without the STRONG operator.

- (25) a. $\forall x[\text{student}(x) \rightarrow [\text{sang}(x) \vee \text{danced}(x)]]$
b. $\text{MOST } x[\text{students}(X) \wedge x \text{ is-part-of } X][\text{sang}(x) \vee \text{danced}(x)]$

This shows that the operator STRONG is arbitrarily introduced to deal with a small portion of data. It does not capture a more general phenomenon including FCEs.

One thing we can learn from Klinedinst(2006) is that when FCEs arise, there is more than one entity involved in the situation. But the relevant NP does not have to be in plural form.

So far I have discussed three previous analyses of FCEs and found the following properties of them:

1. In a disjunction structure, a disjunct can contribute to the meaning of the whole sentence, independently of the other disjunct(s).
2. FCEs also arise in non-monotonic contexts, in which SIs do not.
3. FCEs arise locally.
4. A FCE involves more than one entity in the relevant situation.

In this paper I will try to propose a new analysis which reflects the properties we have seen so far.

3. FCEs as implicatures of existence

We have seen that a FCE arises when a disjunction structure occurs in the scope of an existential quantifier. But as we have seen, similar inferences obtain when a disjunction structure occurs in the scope of other quantifiers.

(26) Some/Many/All students sang or danced.

(27) Some students sang (but did not dance) and some students danced (but did not sing).

Thus the general pattern of FCEs should be like the following:

(28) ... QUANT ... A or B ...

a. ... SOME ... A but not B;

b. ... SOME ... B but not A

FCEs are inferences that arise from the felicity condition that there must be some semantic entities which can be attributed only to one disjunct within the scope of a quantifier. When the quantifier is an existential quantifier, the interpretations we get are called FC interpretations. However, since we get the same inferences from other quantifiers, FCEs are just part of a more general phenomenon. I will call these inferences

implicatures of existence, because they are characterized as existence of some semantic entities.

What does the meaning of existence imply? In relation to this, Klindinst's analysis tells us something. When a quantifier has a disjunction structure, the relevant set of entities has to satisfy some compositional condition: the entities must not have the property due to only one disjunct. Consider (26) again. A set of students is relevant who sang or danced. And the condition on the composition of the students is that not all students sang and that not all students danced. This means that among the students, there are some students who sang but did not dance and some who danced but did not sing, which yields FECs:

- (29) Some of the students who sang or danced ONLY sang.
Some of the students who sang or danced ONLY danced.

This means that either of the two disjuncts is not trivial, but contributes non-trivially to the interpretation of a quantifier. Here the operator ONLY is interpreted with respect to a set of disjuncts. With respect to {A, B, C}, 'ONLY A' means that A, but not B and not C. The sentences in (26) are verified by a set of students who sang or danced. Among the students, some students only sang and others only danced. There must be some entities as the exclusive meaning contribution of each disjunct.

Such inferences arise only when there are students who sang or danced. The following sentence does not give rise to such an inference.

- (30) No students sang or danced.
+/> Some of the students who sang or danced ONLY sang.
+/> Some of the students who sang or danced ONLY danced.

Since there was no student who sang or danced, there is no implicature of existence. The quantifier no students introduces a DE context, but this does not mean that FCEs do not arise in downward-entailing contexts. The quantifier less than ten students also introduces a DE nuclear scope, but we can get the same implicatures:

(31) Less than ten students sang or danced.

+> Some of the students who sang or danced only sang.

+> Some of the students who sang or danced only danced.

This shows that FCEs are restrictions on the composition of semantic entities that verify the quantifier interpretation.

4. FCE: a non-syntactic phenomenon

To account for FCEs, Fox(2006) proposed a syntactic analysis, which assumes a covert *exh*(austive) operator in a syntactic structure. However, this is not supported. (32) can be interpreted in two ways:

(32) John may eat an apple or a pear.

a) John may eat an apple and John may eat a pear.

b) It is possible that John may eat an apple and it is possible that John may eat a pear (but I don't know which).

Both readings come from the use of the disjunction operator and they both have a conjunctive interpretation with existential force. In (32), the auxiliary verb *may* is an existential quantifier over deontic alternatives, while *it is possible that* is an existential quantifier over epistemic alternatives.⁵⁾ Only the first interpretation is considered a FC interpretation, and Fox's analysis only focused on this.

In my analysis, both have the same implicature of existence, as follows:

(33) a. In some of the deontic alternatives of *w** in which John eats an apple or a pear, John ONLY eats an apple;

b. In some of the deontic alternatives of *w** in which John eats an apple or a pear, John ONLY eats a pear.

5) For this reason, Zimmermann(2000) and Geurts(2005) assumes that the disjunction operator is inherently epistemic.

- (34) a. In some of the speaker's epistemic alternatives of w^* in which John may eat an apple or a pear, John might ONLY be allowed to eat an apple;
 b. In some of the speaker's epistemic alternatives of w^* in which John may eat an apple or a pear, John might ONLY be allowed to eat a pear.

Both interpretations involve an existential quantification, but it simply applies to a different domains of quantification, depending on the scope of the disjunction operator. In the second interpretation, the relevant quantifier is a universal quantifier over a set of epistemic alternatives, which is not available in a syntactic structure.

Gazdar(1979) and Sauerland(2004) consider the operator(s) K (and P) in calculating implicatures. They are universal and existential quantifiers over epistemic alternatives for the speaker or for the conversational participants. They are involved in the process of making inference:⁶⁾

- (35) $\parallel K_a(\text{John may eat an apple or a pear}) \parallel^{w^*, u, a}$
 = for every $w \in \text{ALT}(a, w^*)$, John may_w eat an apple or a pear.

Suppose that the sentence is uttered by a speaker a at an utterance time u in the actual world w^* . In the truth-conditions, the relevant set is the set of epistemic alternatives w (of the actual world w^*) in which John may_w eat an apple or a pear. From that set, we can get the implicatures we want, as given in (34). Note that the operator K is introduced pragmatically, not structurally. This shows that the FCE is not a structural matter, but a pragmatic matter.

The K operator is available for all statements, and we can always get FCEs from epistemic alternatives, even when there is no modal operator:

6) First, a statement is considered with respect to the epistemic alternatives for the speaker, which is argued for in Yeom(1997), and reason about his or her information state. But when the hearer gets an implicature, he or she accepts it and changes his or her own information state. Consequently all the conversational participants are considered in calculating implicatures.

(36) John is watching TV or playing games.

+> John might be watching TV.

+> John might be playing games.

If we did not assume an operator like *K*, the implicatures with the meaning of epistemic possibility would be surprising. In this regard, the assumption of *K* is well-motivated.

5. FCEs as meaning contributions of disjuncts

We have the generalization of FCEs as restrictions on the composition of semantic entities in the relevant set (= RS) for a disjunction structure. In this section, I will explain how the idea can be implemented. First, I assume standard interpretation rules because we do not rely on any specific rules of interpretation. And we calculate implicatures from the truth-conditions of a sentence. For a FCE, we get the relevant set of semantic entities for the proper use of a disjunction structure.

(37) a. $\|Q_i[R] [N1 \text{ or } N2] \|_{w,g}$ (*R*: restrictor, *N1* or *N2*: nuclear scope)

the RS in $w = \{d \in D: \|R \|_{w,g[i/d]} \wedge [\|N1 \|_{w,g[i/d]} \vee \|N2 \|_{w,g[i/d]}]\}$

Implicatures:

i. $\{d \in D: \|R \|_{w,g[i/d]} \wedge \|N1 \|_{w,g[i/d]} = 1 \wedge \|N2 \|_{w,g[i/d]} = 0\} \neq \emptyset$

ii. $\{d \in D: \|R \|_{w,g[i/d]} \wedge \|N1 \|_{w,g[i/d]} = 0 \wedge \|N2 \|_{w,g[i/d]} = 1\} \neq \emptyset$

b. $\|M[N1 \text{ or } N2] \|_{w,g}$ (*M*: a modal operator)

$= \|M \|_{w,g}(\lambda w' [w'Rw \wedge [\|N1 \|_{w',g} \vee \|N2 \|_{w',g}]])$

(“*w'Rw*” means that *w'* is an accessible alternative of *w* with respect to the modality determined by the context.)

the RS in $w = \{w' \in W: w'Rw \wedge [\|N1 \|_{w',g} \vee \|N2 \|_{w',g}]\}$

Implicatures:

i. $\{w' \in W: w'Rw \wedge \|N1 \|_{w',g} = 1 \wedge \|N2 \|_{w',g} = 0\} \neq \emptyset$

ii. $\{w' \in W: w'Rw \wedge \|N1 \|_{w',g} = 0 \wedge \|N2 \|_{w',g} = 1\} \neq \emptyset$

Here (a) is a case where the quantifier quantifies over individuals, whereas (b) is a case where the quantifier quantifies over possible worlds. $w'Rw$ is an accessibility relation between w' and w and it is determined by the context, whether it is a deontic or epistemic one. Thus, if a quantifier is used in a complement clause of a verb like think, the interpretation is like the following:

- (38) $\parallel$ John thinks some students sang or danced $\parallel^{w*,g}$
 = for every $w \in ACCE(j, w^*)$, $\parallel$ some students sang or danced $\parallel^{w,g}$

Here w is an epistemic alternative of w^* for John and the embedded clause is interpreted with respect to w and g , which are determined from the main context and the semantics of the verb think.

Once a local context w and g are determined, we can get the RS for the disjunction structure. From the RS, we can get the implicature that each disjunct makes an exclusive meaning contribution to the composition of the RS of semantic entities. That is, that the set of semantic entities that make only one disjunct hold should not be the empty set.

6. FC Effects as a Global Phenomenon

At first we gave the general pattern of getting FCEs, as given in (4). This means that FCEs can occur locally and they simply combine with the meaning of the rest of a sentence. This predicts the following:

- (39) John believes that some students sang or danced.
 +> John believes that some of the students who sang or danced sang (but did not dance)
 +> John believes that some of the students who sang or danced danced (but did not sing)

FCEs arise in the embedded clause and they combine with the meaning of "John believes that"

From this we can predict that no FCEs arise when the FCEs arise locally in a DE context.

(40) John does not believe that some students sang or danced.

+> John does not believe that some/any of the students who sang or danced sang (but did not dance)

+> John does not believe that some/any of the students who sang or danced danced (but did not sing)

This is simply because John does not believe that there are students who sang or danced. That is, John believes that no students sang or danced, and no FCEs arise for the same reason that no FCEs arise in (30).

However, we cannot say in general that FCEs simply combine with the meaning of the rest. Consider the following example.

(41) John does not believe that many students sang or danced.

a. ??John does not believe that some/any of the students who sang or danced only sang. (= John believes that none of the students who sang or danced only sang.)

b. ??John does not believe that some/any of the students who sang or danced only danced. (= John believes that none of the students who sang or danced only danced.)

c. John believes that (it is possible that) some of the students who sang or danced only sang.

d. John believes that (it is possible that) some of the students who sang or danced only danced.

In this example, the quantifier many students is used instead of some students. When John does not believe that many students sang or danced, it is compatible with the proposition that John believes that some students sang or danced. This allows us to get the RS of students who sang or danced in some or all of John's belief worlds. In such belief worlds, we get FCEs. For this reason, (41a) and (41b) are unacceptable because the FCEs are completely negated. This yields the effects of (41c) and (41d). If

the FCEs arise in some of John's belief worlds, they come with the possibility operator it is possible that. If they arise in all of John's belief worlds, we get stronger FCEs without the possibility operator.

When other quantifiers than existential quantifiers are involved, we do not call the implicatures of existence FCEs. But the effects are the same, and they arise from the same reasoning. Therefore we do not have to distinguish them. When a disjunction structure lies in the scope of a downward monotonic quantifier, no FCEs arise only because no RS of semantic entities is formed for implicatures of existence.

7. Conclusion

In this paper I discussed previous analyses on FCEs critically, trying to find out what features they capture and what problems they have. One crucial feature is that in a disjunction structure, each disjunct can be compositionally interpreted with the rest of the sentence, independently of the other disjunct(s). This semantic property is combined with the fact that a disjunction structure is more complex than the corresponding sentence only with one disjunct. Thus the use of a disjunct implicates that there is some meaning contribution of each disjunct to the meaning of the whole sentence. This is the basic idea. However, meaning contributions are not made directly to the meaning of the whole sentence, but to the interpretation of a quantifier which has immediately wide scope over a disjunction structure. And the meaning contribution from the quantifier to the whole sentence is done in a separate process. For this reason, the meaning contribution should be taken to be an implicature: an implicature arises from an utterance as a whole, and each disjunct should contribute ultimately to the meaning of the whole sentence.

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